

1. PROJECT PLAN AND CURRENT IMPLEMENTATION

This report presents the project as planned around the current VIVE architecture and summarizes the implementation achievements recorded so far, followed by the planned integration/optimization work still in the roadmap.

2. WHAT WAS PLANNED

Area	Planned implementation
Foundation	Repository, contracts, backend, authentication and session management.
Streaming	WebSocket, ring buffer, VAD, packetization, audio quality and packet traceability.
AI layer	Silero VAD, AASIST, ECAPA-TDNN, Indic ASR, intent and behavior models.
Intelligence	Context, OOD/uncertainty, risk fusion, temporal risk and policy engine.
Product	Android application, dashboard, packet forensics, alerts, reports and notifications.
Integration	REST/WebSocket APIs, SDK boundary, authorized cellular screening and authorized full-audio paths.
Validation	Component evaluation, end-to-end testing, robustness, calibration, latency, security and privacy.
Deployment	Model/version tracking, monitoring, reproducible configuration and production-oriented service boundaries.

3. ACHIEVED IN THE CURRENT VIVE IMPLEMENTATION

- FastAPI backend and WebSocket streaming path implemented.
- Android client connected to the live session/backend path.
- Real microphone capture and streaming packetization implemented.
- Sliding 2-second windows with 1-second stride implemented.
- VAD integrated for speech gating.
- AASIST integrated as the anti-spoof evidence model.
- ECAPA-TDNN integrated with explicit NO_REFERENCE behavior when enrollment is unavailable.
- IndicConformer multilingual ASR integrated; real Hindi/Tamil processing was tested.
- Intent classifier integrated with the defined VIVE taxonomy.
- Behavior classifier integrated with the defined VIVE taxonomy.
- Risk fusion and temporal risk logic integrated; risk and confidence remain separate.
- Failure/uncertainty states are represented explicitly rather than turning every failure into high risk.
- Android live session UI shows risk, confidence, duration, packets, language, transcript/evidence and packet progression.
- Alert notification path and backend alert handling implemented.
- Reports/session history and model/status views connected to the real backend path.
- Security/privacy controls, model path configuration and real-model adapters were implemented and audited during the build.
- Real end-to-end device sessions were exercised with actual microphone audio and actual backend models.

4. MODEL/INFERENCE ACHIEVEMENTS

Layer	Current implementation status in project records
VAD	Silero VAD integrated and used for speech gating.
Anti-spoof	AASIST integrated and evaluated. Proper ASVspoof evaluation confirmed the implementation works; target phone/call-like transfer remains an evaluation focus.
Speaker	ECAPA-TDNN integrated; enrollment/reference workflow defined. NO_REFERENCE is returned when a reference is unavailable.
ASR	IndicConformer 600M selected and integrated for Hindi/Tamil/Indic ASR; real Tamil and Hindi streaming tests were performed.
Intent	Project classifier trained and integrated for VIVE intent taxonomy.
Behavior	Project classifier trained and integrated for VIVE behavior taxonomy.
Risk	Transparent multi-signal fusion and temporal logic implemented; calibration/evaluation remains part of the planned validation work.

5. PLANNED NEXT INTEGRATION / OPTIMIZATION WORK

- Complete the mobile-first model strategy so the core analysis path can operate without depending on a laptop backend.
- Select and benchmark a mobile-feasible multilingual ASR stack against the reference ASR quality/latency requirements.
- Complete speaker enrollment/reference persistence and same/different-speaker evaluation.
- Strengthen the anti-spoof component for the actual target phone/call-like audio domain and use only validated evidence in risk fusion.
- Expand legitimate service-call / real-call-like semantic evaluation so intent and behavior do not over-trigger on benign conversations.
- Expand Tamil semantic support and multilingual/codeswitching handling.
- Complete calibration and threshold tuning using the project-defined held-out evaluation protocol.
- Complete robustness evaluation across telephone/VoIP compression, noise, reverberation, short windows and unseen generators.
- Complete final phone-only acceptance measurements: startup, RAM, model size, latency, sustained streaming and battery/resource behavior.
- Complete final integration verification for notifications, reports, APIs and authorized communication paths.

6. CURRENT PROJECT STATE IN ONE VIEW

Layer	State
Architecture	Defined and implemented around the multi-evidence VIVE pipeline.
Backend	Real streaming/inference path implemented.
Android	Live capture/session/risk/evidence/notification workflow implemented.
Real ML	Integrated across VAD, anti-spoof, speaker, ASR, intent and behavior layers.
Evaluation	Component and end-to-end evaluation infrastructure established; further target-domain/mobile evaluation is in the roadmap.
Integration	Authorized cellular screening + authorized full-audio architecture defined.
Production hardening	Security, privacy, monitoring, versioning and failure handling are part of the implementation plan and current codebase.

7. ACCEPTANCE END STATE

The planned final demonstration is:

Authorized voice stream → real-time packets → real model inference → multilingual transcript → intent/behavior → contextual evidence → temporal risk → explainable alert → dashboard + notification/API output.

The final engineering evidence will be presented through component metrics, end-to-end event metrics, latency, robustness, calibration and device measurements rather than a single generic accuracy number.